

SULAYMANIYAH, Iraq

WMO: 406230

Lat: 35.550N

Lon: 45.450E

Elev: 2799

StdP: 13.27

Time Zone: 3.00 (E03)

Period: 89-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.5	(c) 26.9	(d) 10.3	(e) 10.3	(f) 44.7	(g) 14.2	(h) 12.6	(i) 46.7	(j) 23.3	(k) 46.9	(l) 21.1	(m) 48.1	(n) 3.9	(o) 300	(p) 0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 34.2	(c) 109.5	(d) 66.8	(e) 107.5	(f) 66.5	(g) 105.5	(h) 65.7	(i) 70.1	(j) 102.4	(k) 68.5	(l) 100.7	(m) 67.3	(n) 99.6	(o) 9.9	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
60.6	87.7	72.1	58.8	82.2	70.4	57.1	77.1	70.3	35.8	101.7	34.4	100.2	33.3	100.0	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.7	(b) 18.7	(c) 16.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 21.7	(b) 18.7	(c) 16.9	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.7	42.5	44.7	52.0	58.9	69.9	82.2	88.8	88.5	79.5	67.9	54.4	45.6		
	DBStd	17.35	4.98	6.09	5.12	5.53	6.35	5.15	4.22	4.39	5.63	5.89	5.70	5.15		
	HDD50	615	235	164	36	4	0	0	0	0	0	0	23	153		
	HDD65	2844	696	567	405	194	23	0	0	0	1	38	319	601		
	CDD50	5981	4	16	98	271	617	967	1202	1194	885	554	155	18		
	CDD65	2736	0	0	1	12	175	517	737	729	437	127	1	0		
	CDH74	44791	0	0	50	293	2868	8516	12540	12064	6538	1872	44	6		
	CDH80	29322	0	0	6	47	1349	5532	8933	8560	4127	767	1	0		
(14) Wind	WSAvg	6.8	5.7	5.9	6.7	6.3	6.5	8.1	8.2	8.0	7.1	6.7	6.2	5.9		
(15) Precipitation	PrecAvg	30.7	5.0	5.4	4.8	4.4	1.8	0.2	0.1	0.0	0.1	1.2	3.7	4.4		
	PrecMax	44.1	8.4	12.8	10.8	9.1	4.8	0.6	0.2	0.2	0.4	4.9	11.5	11.7		
	PrecMin	17.5	1.5	2.6	0.9	0.5	0.1	0.0	0.0	0.0	0.0	0.0	0.2	0.4		
	PrecStd	7.4	2.0	2.4	2.5	2.4	1.5	0.2	0.1	0.1	0.1	1.3	3.5	3.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.8	69.2	80.6	84.6	97.1	107.5	112.8	111.4	105.8	94.8	78.4	71.8		
		MCWB	48.7	51.6	55.5	62.4	63.2	66.8	67.8	67.5	65.2	62.8	54.9	49.4		
	2%	DB	58.9	64.8	73.5	80.6	94.6	104.3	109.8	109.4	103.6	91.2	75.0	64.2		
		MCWB	47.3	50.3	55.1	59.7	63.0	65.1	66.8	66.7	63.7	61.7	52.7	47.4		
	5%	DB	56.6	61.1	69.7	77.2	91.4	102.3	107.8	107.5	100.5	88.0	71.8	60.1		
		MCWB	46.8	48.5	52.8	59.1	62.0	64.6	66.4	66.8	63.6	60.0	52.6	47.6		
	10%	DB	53.8	57.5	66.2	73.5	88.1	100.0	105.7	105.7	97.4	84.5	68.1	56.8		
		MCWB	45.4	47.4	52.2	58.3	61.4	63.9	65.3	66.3	62.8	58.9	51.8	47.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	55.0	60.0	65.7	68.5	69.8	72.5	72.3	69.1	66.5	60.8	54.5		
		MCDB	56.9	62.8	71.1	79.4	84.4	101.4	106.5	105.9	98.9	89.0	70.9	58.4		
	2%	WB	49.9	52.6	57.2	63.2	66.4	67.7	69.8	70.4	66.5	63.8	57.8	52.1		
		MCDB	55.9	59.2	70.0	74.8	85.5	99.7	103.7	103.7	96.0	84.5	65.4	57.5		
	5%	WB	48.4	50.4	55.3	61.2	64.8	66.2	68.2	68.8	65.1	62.0	55.9	50.2		
		MCDB	53.4	57.7	65.0	72.8	84.2	98.6	102.1	102.3	95.4	80.4	64.2	55.5		
	10%	WB	47.0	48.7	53.6	59.5	63.2	64.8	66.9	67.3	63.8	60.6	54.1	48.4		
		MCDB	51.8	56.0	63.0	69.3	82.7	96.5	101.3	101.6	94.5	78.6	62.7	54.3		
(35) Mean Daily Temperature Range	MDBR		21.0	22.6	24.4	25.9	30.8	33.9	34.2	34.9	35.0	29.9	25.4	20.2		
	5% DB	MCDBR	26.4	29.4	32.5	32.8	36.1	37.0	37.0	36.4	37.1	35.5	32.4	27.6		
		MCWBR	16.6	17.2	16.8	15.9	13.8	14.8	14.6	15.2	15.5	17.0	16.9	16.3		
	5% WB	MCDBR	23.0	25.1	27.1	27.5	32.0	34.9	33.0	34.5	34.7	30.3	24.2	20.8		
		MCWBR	15.6	16.0	15.3	14.6	14.1	14.9	14.3	15.2	15.7	15.6	13.9	13.8		
(40) Clear-Sky Solar Irradiance	taub		0.370	0.408	0.458	0.540	0.533	0.470	0.499	0.461	0.419	0.443	0.382	0.365		
	taud		2.147	2.046	1.933	1.773	1.808	1.956	1.891	1.985	2.079	2.005	2.162	2.175		
	Ebn at Noon		254	258	255	241	245	260	252	259	264	242	247	247		
	Edh at Noon		37	46	56	69	68	59	62	56	48	47	36	34		
(44) All-Sky Solar Radiation	RadAvg		780	1029	1433	1753	2161	2574	2456	2244	1879	1333	962	738		
	RadStd		71	104	137	154	137	119	61	60	89	100	108	94		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page